Introduction to Segmentation

Thresholding in FIJI: Image Processing, Step 1

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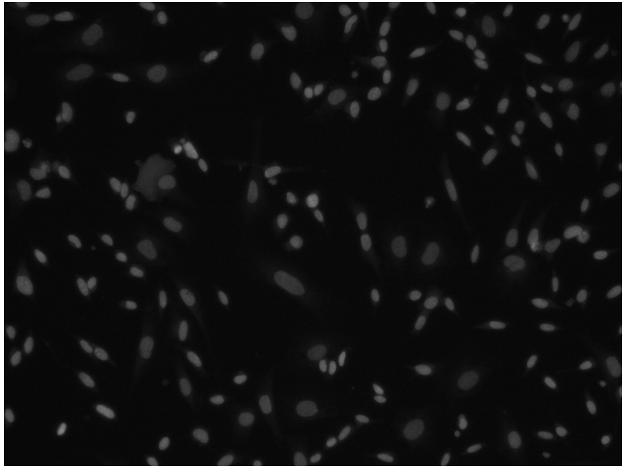
BONN

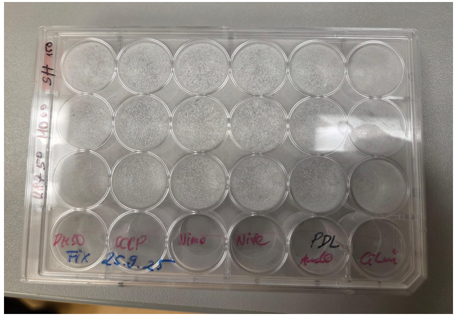
2D Fluorescence Microscopy, Workflow

Image Acquisition



Samples → Images → Data → Data Analysis

Image Processing



Batch	Group	Condition	Total thresholded area	Mean thresholded area
2026	Astrocytes	Ctrl	6550.774	3275.387
2026	Astrocytes	Nimo	6062.291	3031.145
2026	Oligos	Ctrl	5594.899	2797.449
2026	Oligos	Nimo	5213.5	2606.75
2025	Astrocytes	Ctrl	4890.56	2445.28
2025	Astrocytes	Nimo	4608.927	2304.464
2025	Oligos	Ctrl	4365.789	2182.894
2025	Oligos	Nimo	4141.386	2070.693

Image Processing

Despite the variety of workflows and scenarios, the first, or one of the first steps in image processing is segmentation

Segmentation: separation of cellular objects from the background into groups

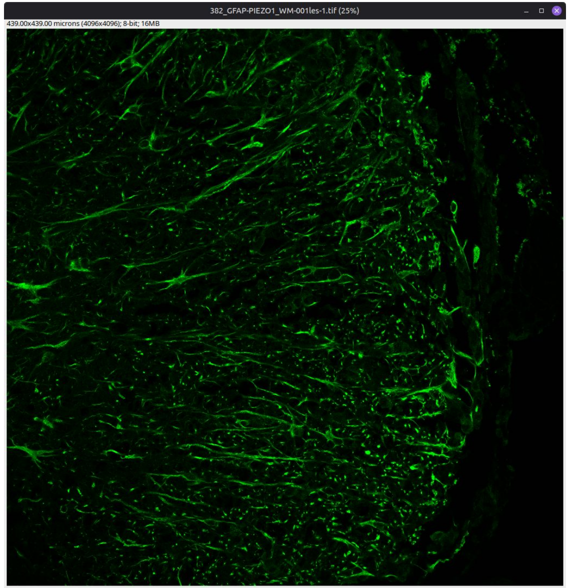
There are different types of (automatic) segmentation algorithms, each of which is based on a particular principle.

In contrast, manual segmentation refers to manual outlining of cellular objects of interest from the background. When comparing against manual segmentation, various segmentation algorithms yield variable and somewhat inaccurate results, which can impact downstream processing. [Ref. 2]

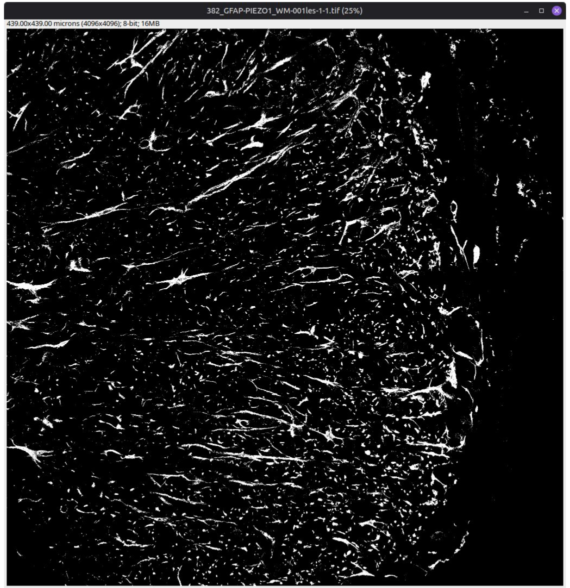
As it turns out, the method oftentimes closest to manual segmentation was a single global threshold determined in advance by the expert.

Thresholding Assistant: a Fiji Macro to support the user in determining this global threshold value

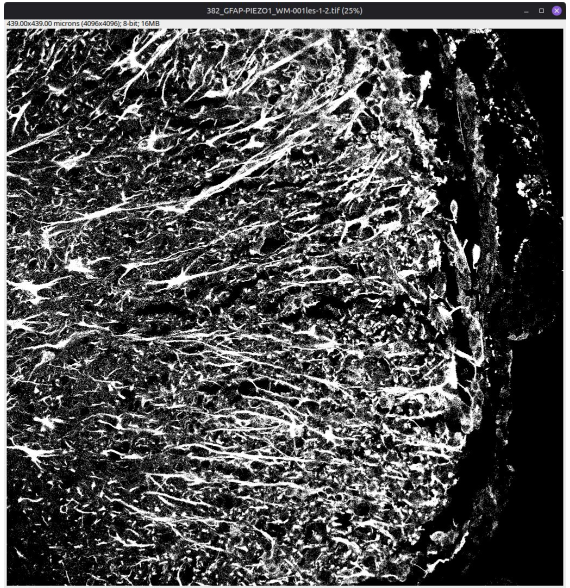
Automatic Segmentation in Action 1/2



Original

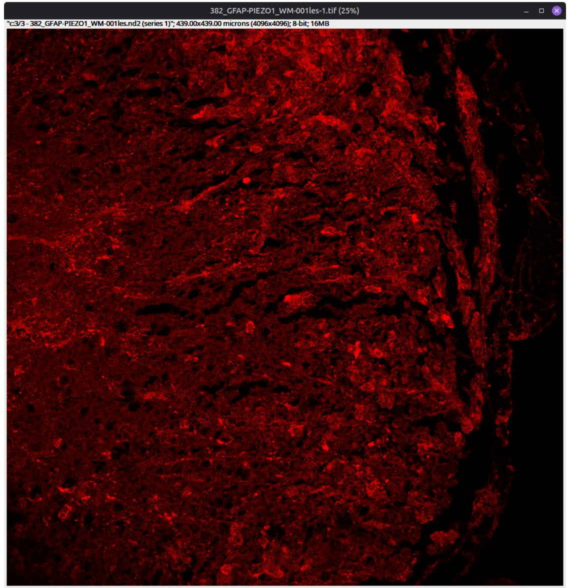


Otsu

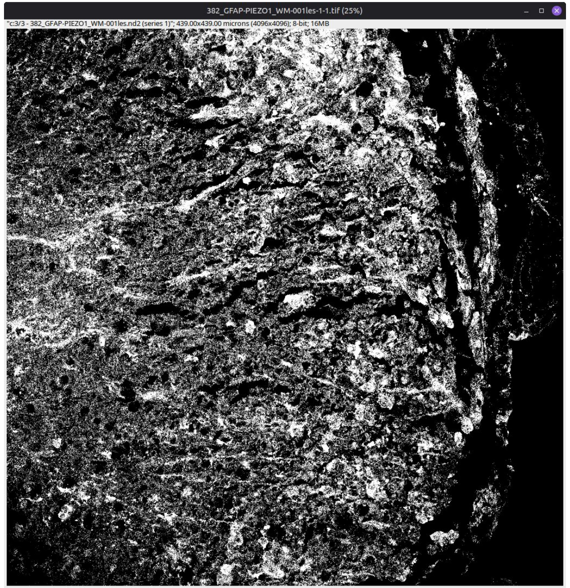


Huang

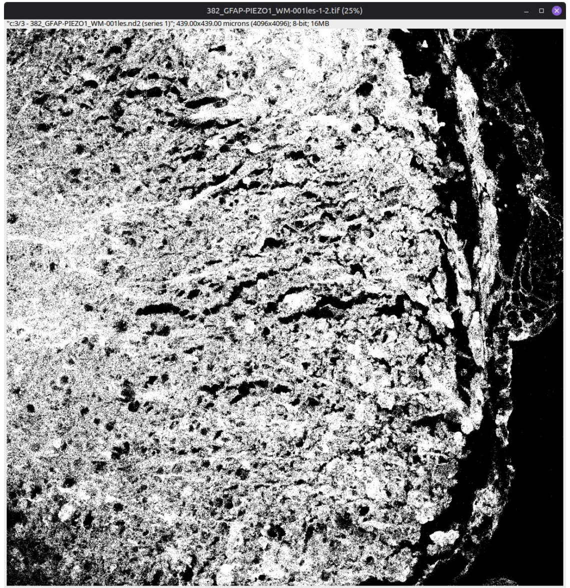
Automatic Segmentation in Action 2/2



Original



Otsu



Huang

Fiji and the “Thresholding Assistant” Macro

What is Fiji?

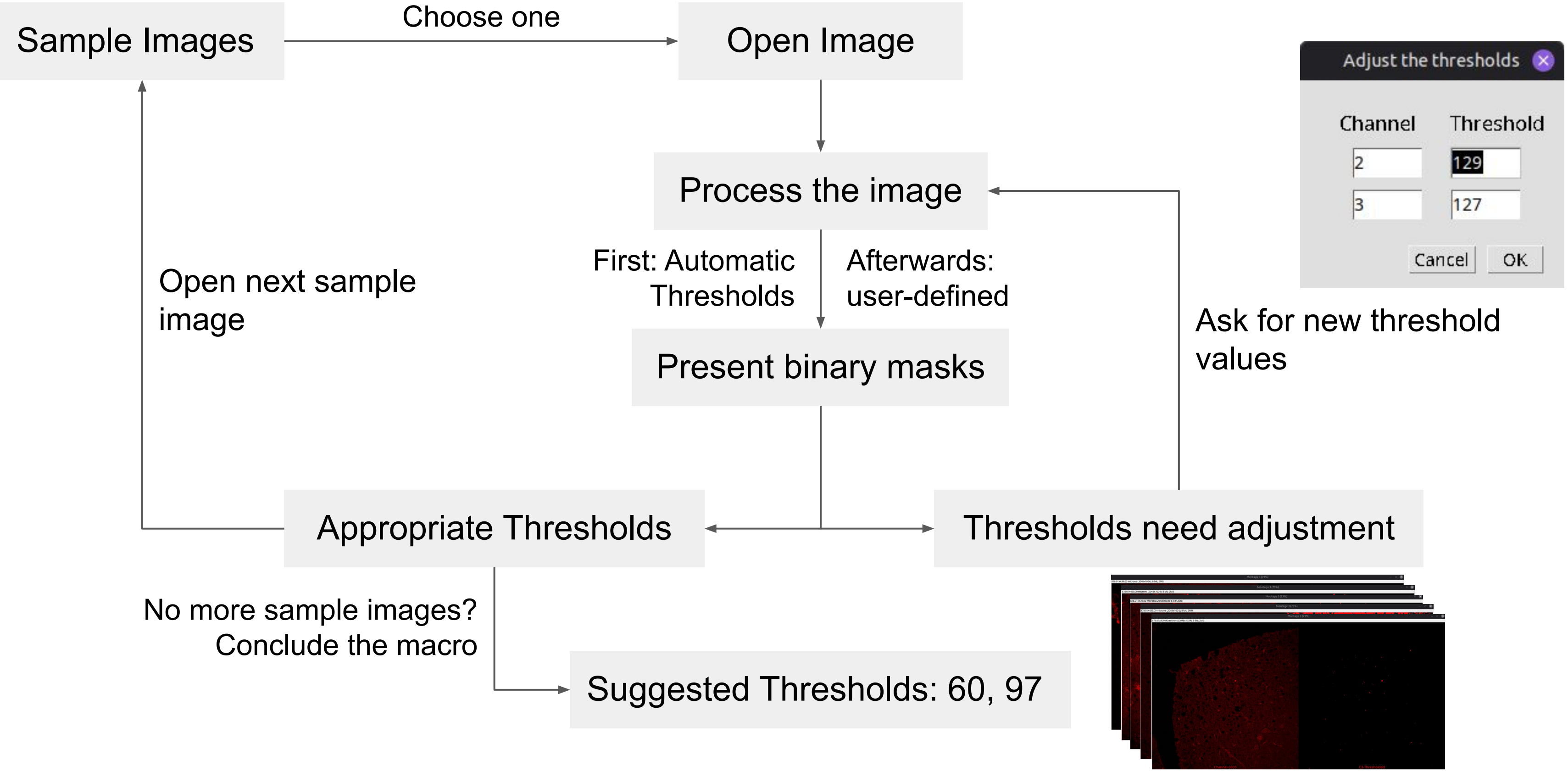
Fiji is an image processing package. Why use Fiji?

1. Free and open source
2. Easy to install and use
3. Available on most operating systems
4. Bundled with a lot of preinstalled plugins
5. Has extensive documentation

Why use this script?

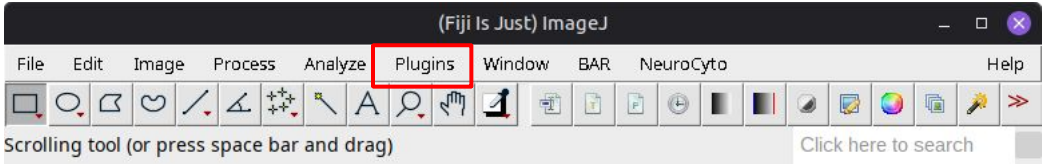
1. It streamlines the discovery of valid threshold values
2. The macro is accessible and easy to use
3. May help to find the proper balance between time and accuracy

Thresholding Assistant: Flowchart



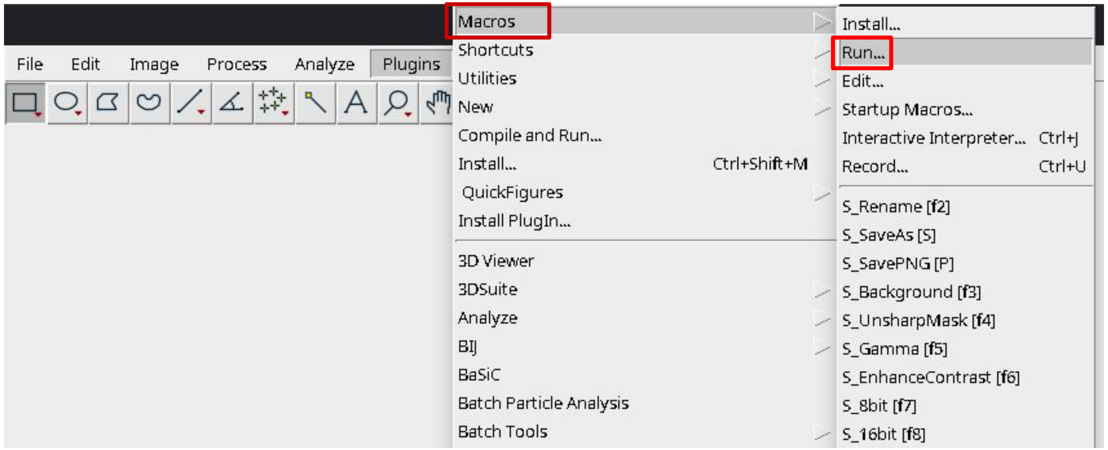
How to run the macro 1/2

1. Download the macro
2. Open Fiji
3. Press “Plugins” (in red)
4. Click on “Macros”
5. Press “Run”
6. Select the macro in the pop-up window



How to run the macro 2/2

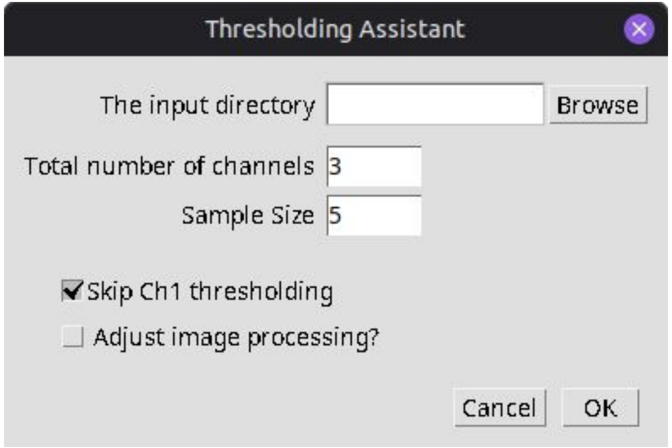
1. Download it
2. Open Fiji
3. Press “Plugins”
4. Click on “Macros”
5. Press “Run”
6. Select the macro in the pop-up window



After that, the macro will run and the first dialog window will appear (see the next slide)

Dialog 1: Basic setup

- Select folder with .tif images (input directory).
- Provide the number of channels in the images. This value is set to 3 by default.
- Sample Size: how many randomly selected images would be representative of the entire collection of images? This value is set to 5 by default, but it is recommended to increase it to 7-10.
- Check the “Skip Ch1 thresholding” box if it is not necessary to estimate Ch1 threshold (Channel 1 being a reference/structural channel)
- “Adjust Image Processing?” needs to be chosen to alter the processing pipeline!



Dialog 2: *“Adjust Image Processing?”*

Image Processing after Thresholding

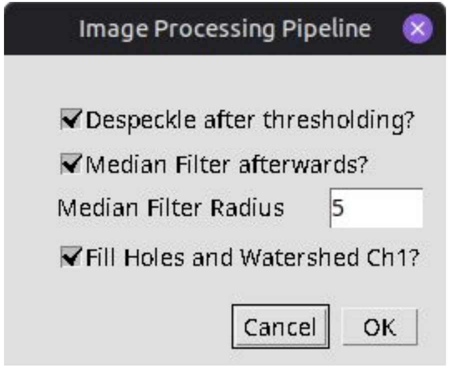
The built-in image processing workflow consists of the following steps:
Threshold -> Despeckle -> Median Filter -> Fill Holes -> Watershed

Each of these operations is *optional* (except thresholding) - see the following slides for illustrations

1. Threshold: segments the image into foreground and background
2. Despeckle: removes small particles from the foreground
3. Median Filter: removes noise and preserves edges

Channel 1-specific operations:

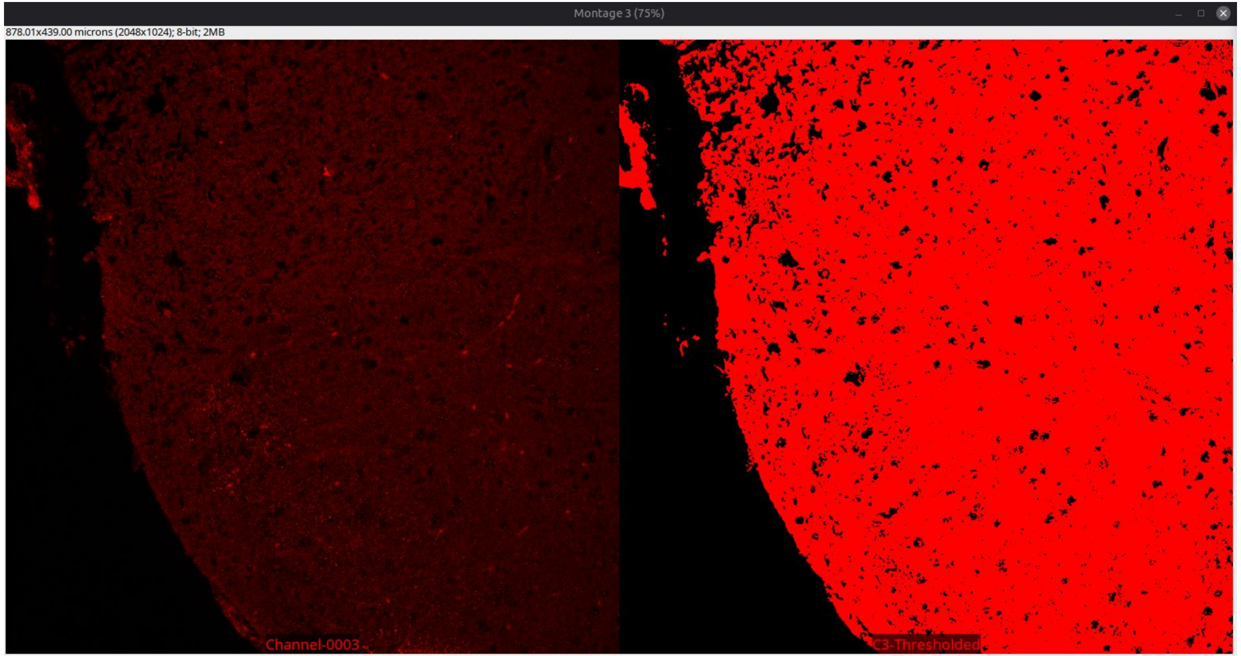
4. Fill Holes: if there is a ring-like particle, e.g. whose rim pixels have higher intensity than the center pixels, then everything within the ring will be also be counted as particle
5. Watershed: separates aggregated, clumped particles



Click on “OK” to confirm the settings and start estimating the thresholds!

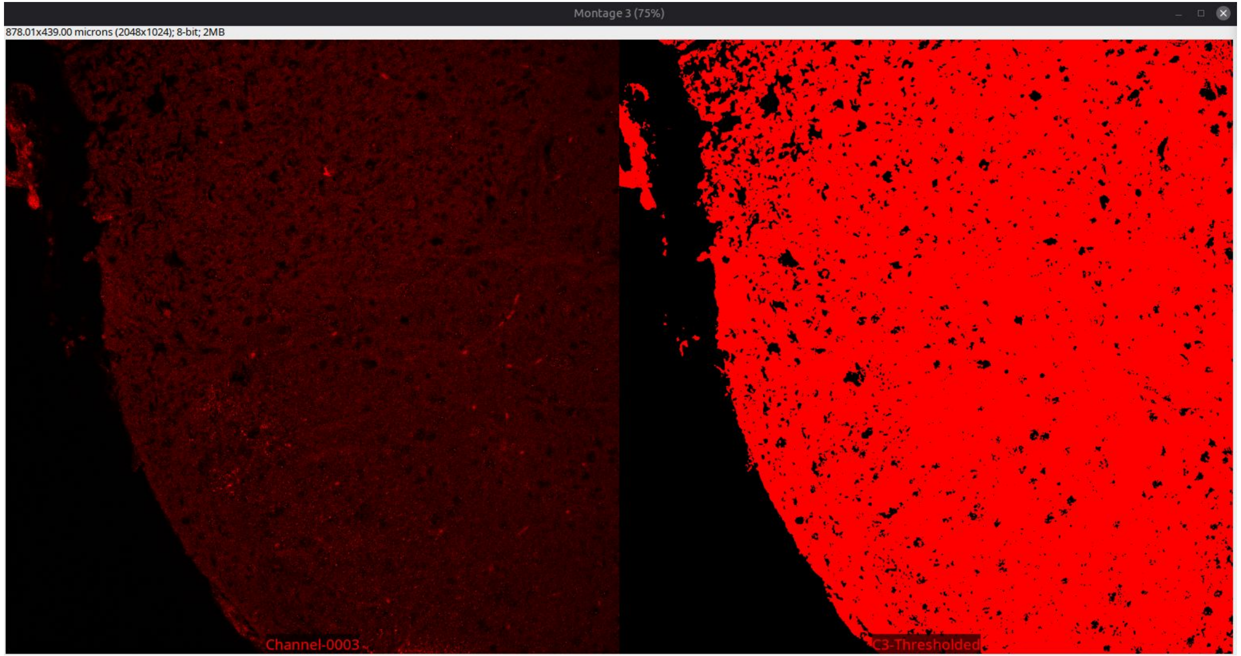
Macro 3/6

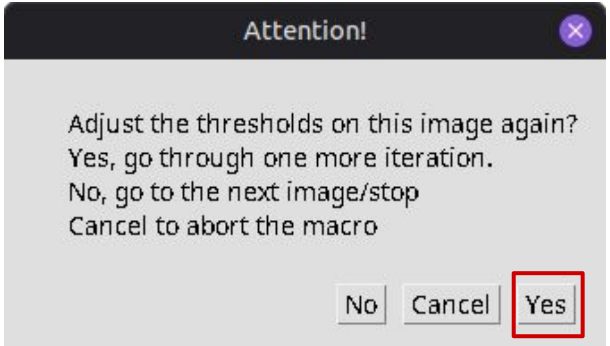
1. Inspect the automatically estimated threshold - it might be needed to move the “Montage” windows to view each channel independently



Macro 3/6

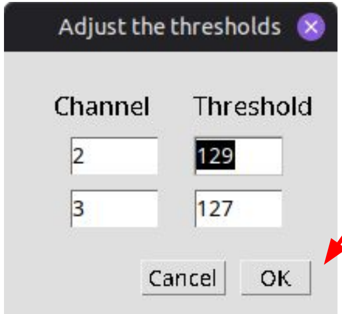
1. Inspect the automatically estimated threshold - it might be needed to move the “Montage” windows to view each channel independently



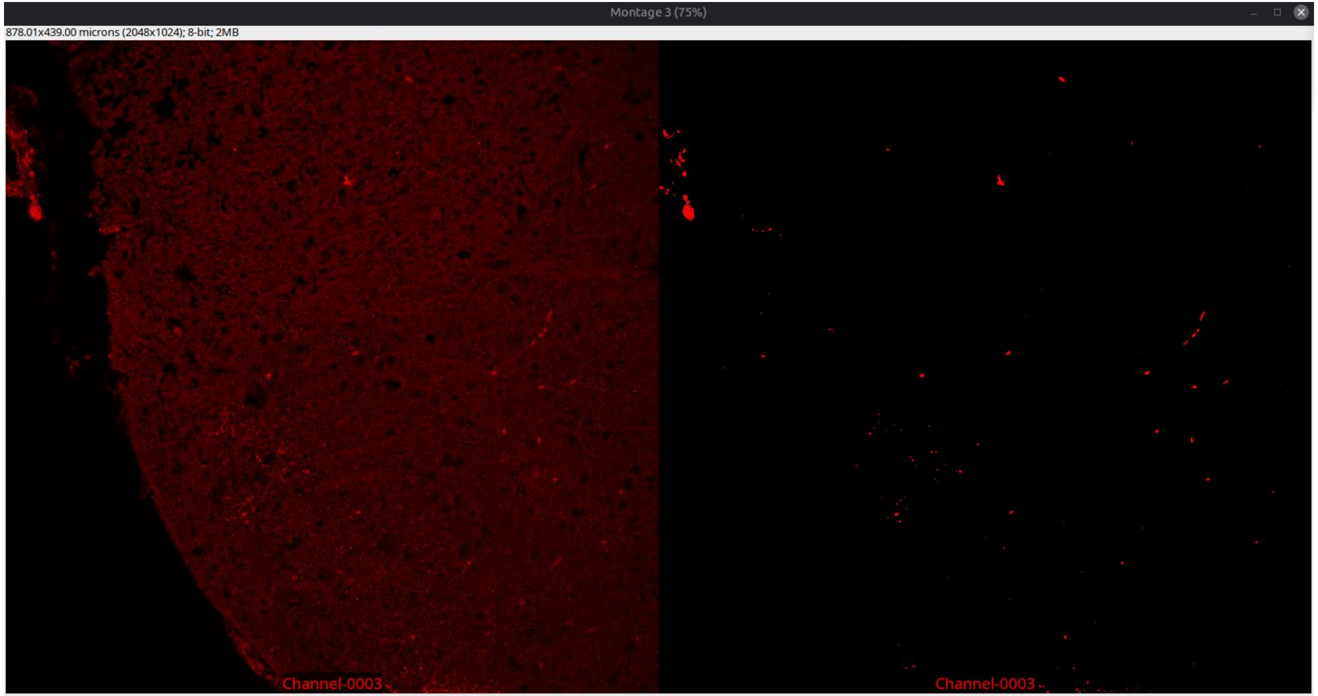


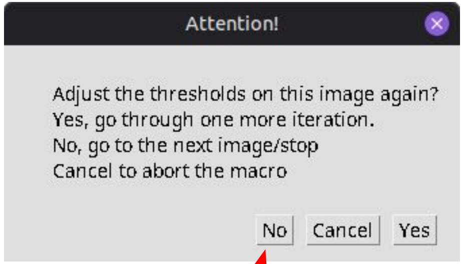
2. If it seems necessary to adjust the threshold, click “Yes”

Macro 4/6



3. Change the values accordingly and click “OK”

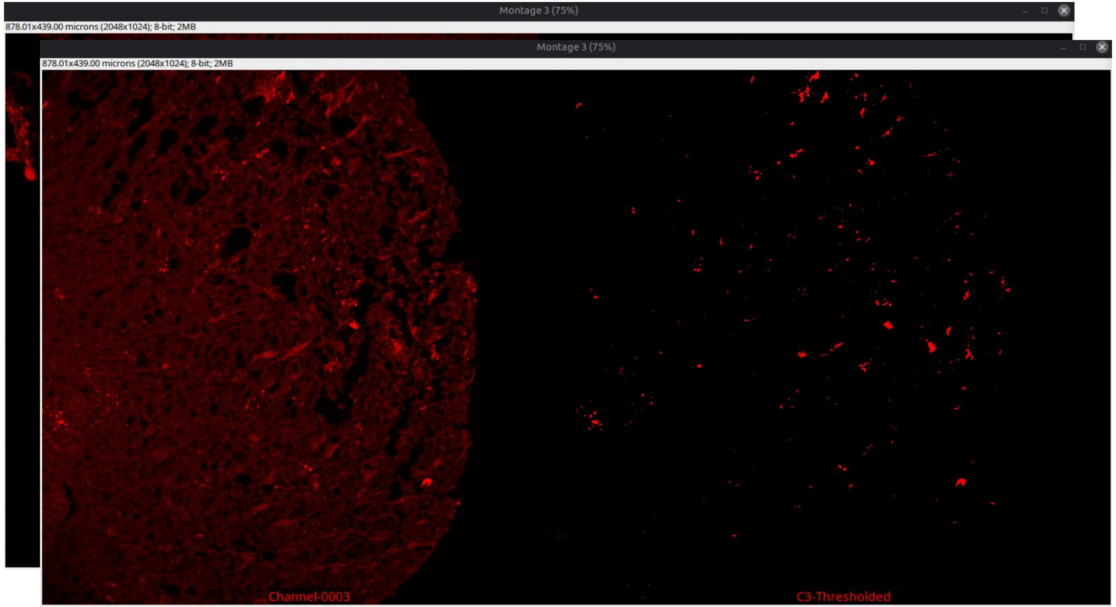


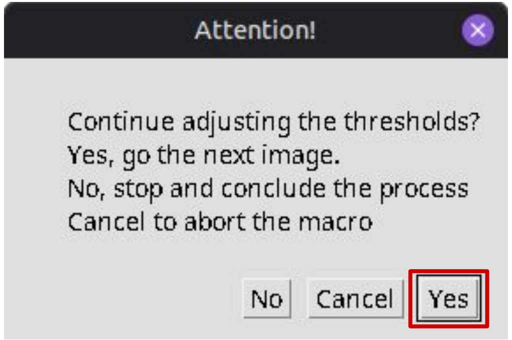


5. Click on “Yes” to iterate one more time, or click on “No” and move on to the next sample image

4. Inspect the new estimate

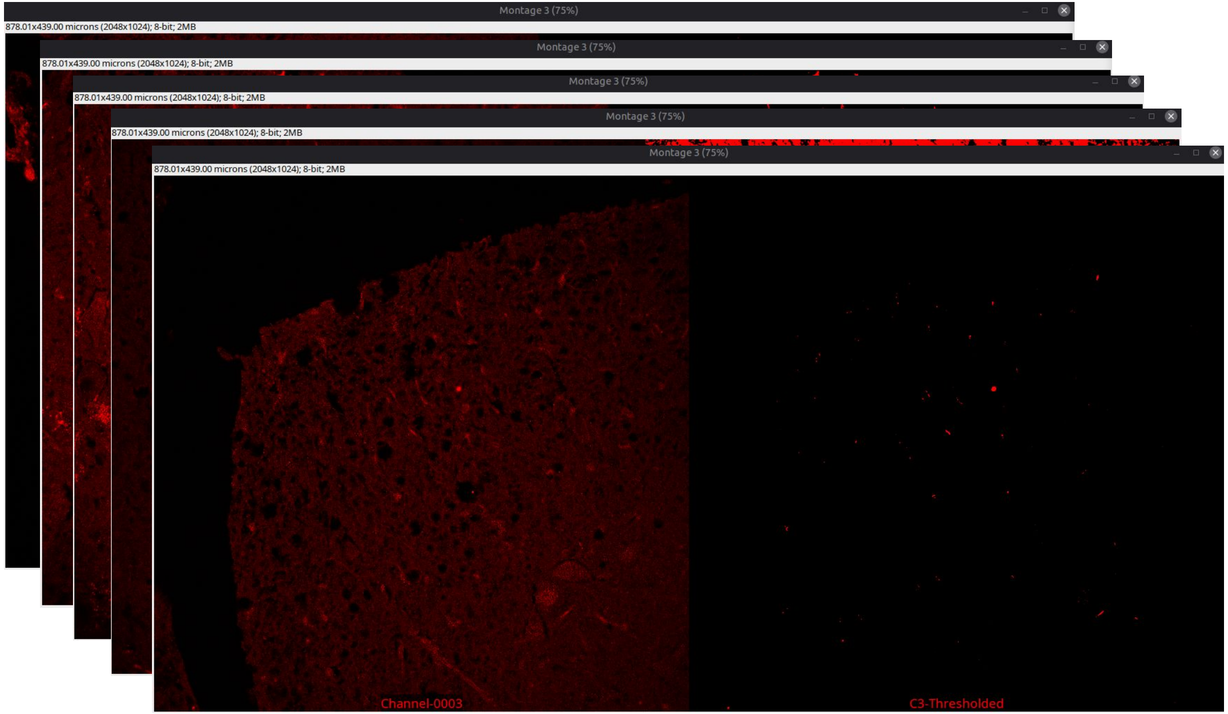
Macro 5/6

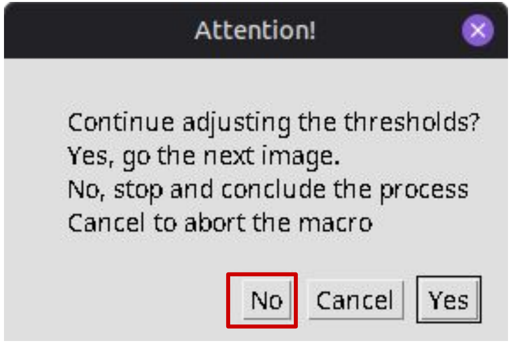




As the number of sampled images is still low, it is reasonable to inspect more images

Macro 6/6





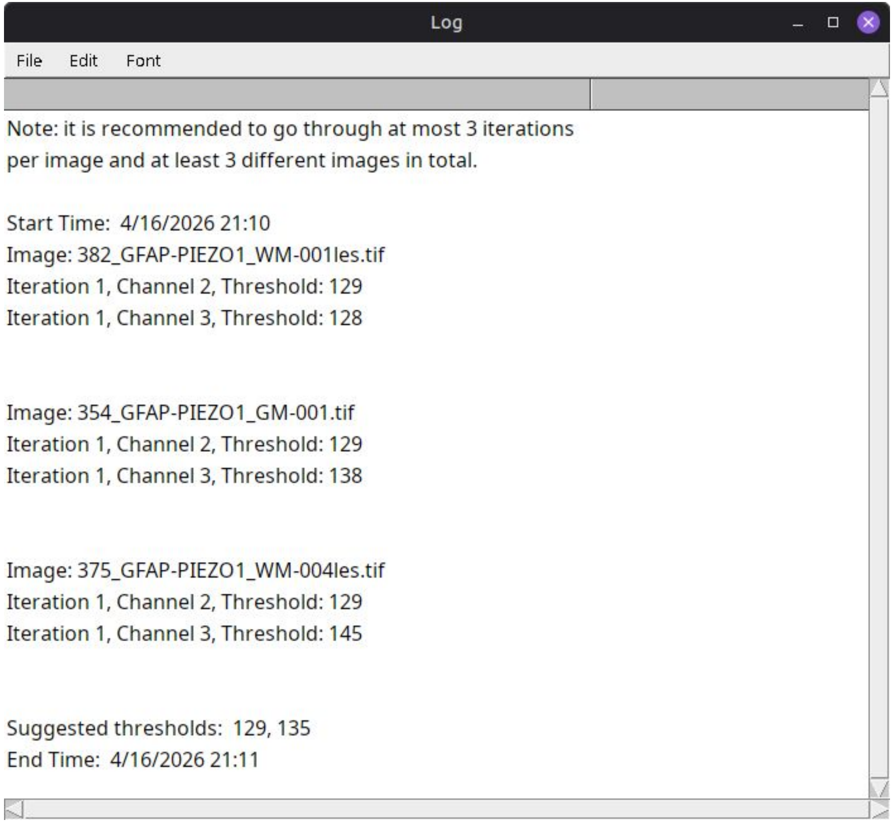
After being done with estimating the thresholds, click on “No” in the pop-up window

Session Information

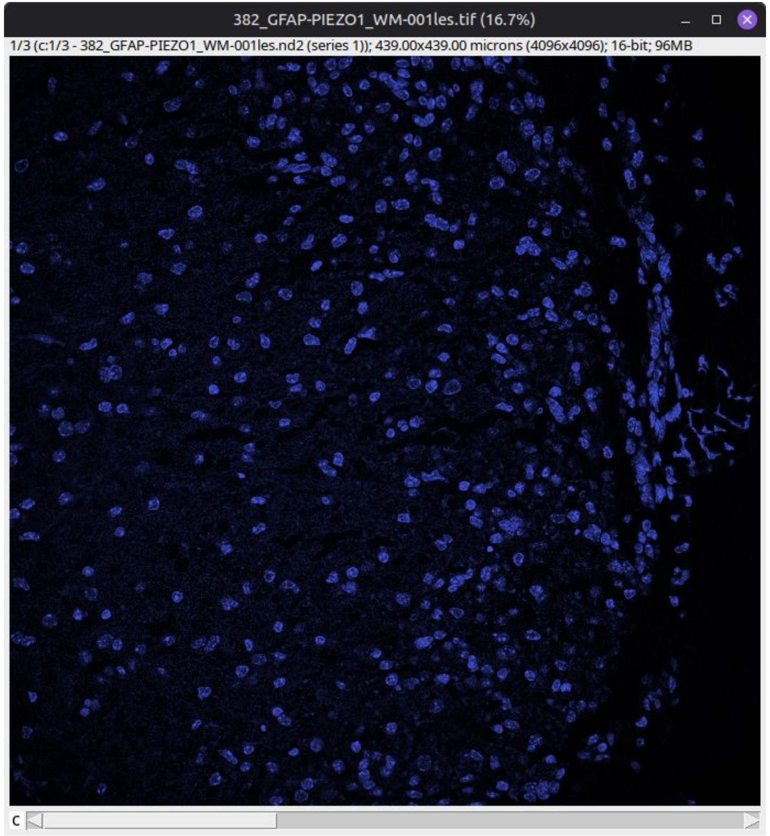
Saved as a .txt file to the output folder

1) Timestamps
2) Images
3) Iterations
4) Channels
5) Thresholds
6) Suggested thresholds

For easier tracking, troubleshooting and reporting

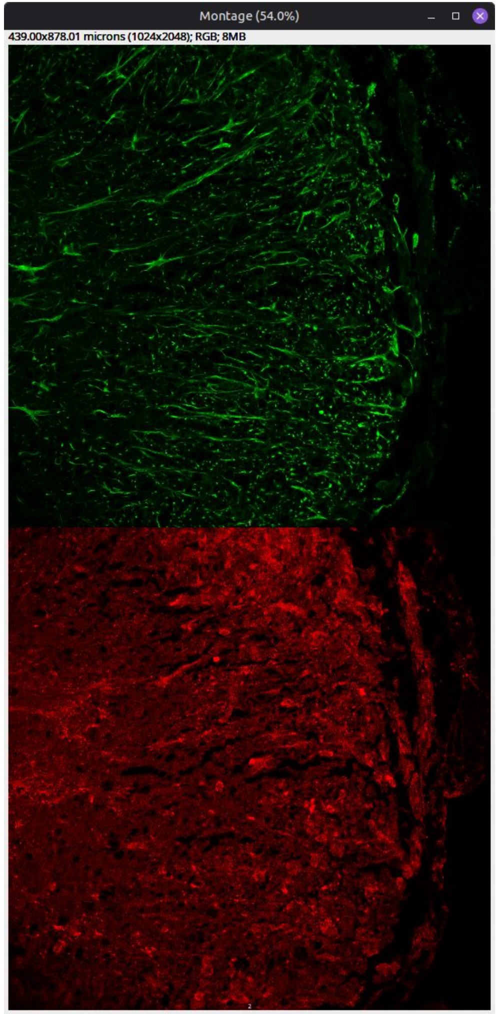


Processing 1/5

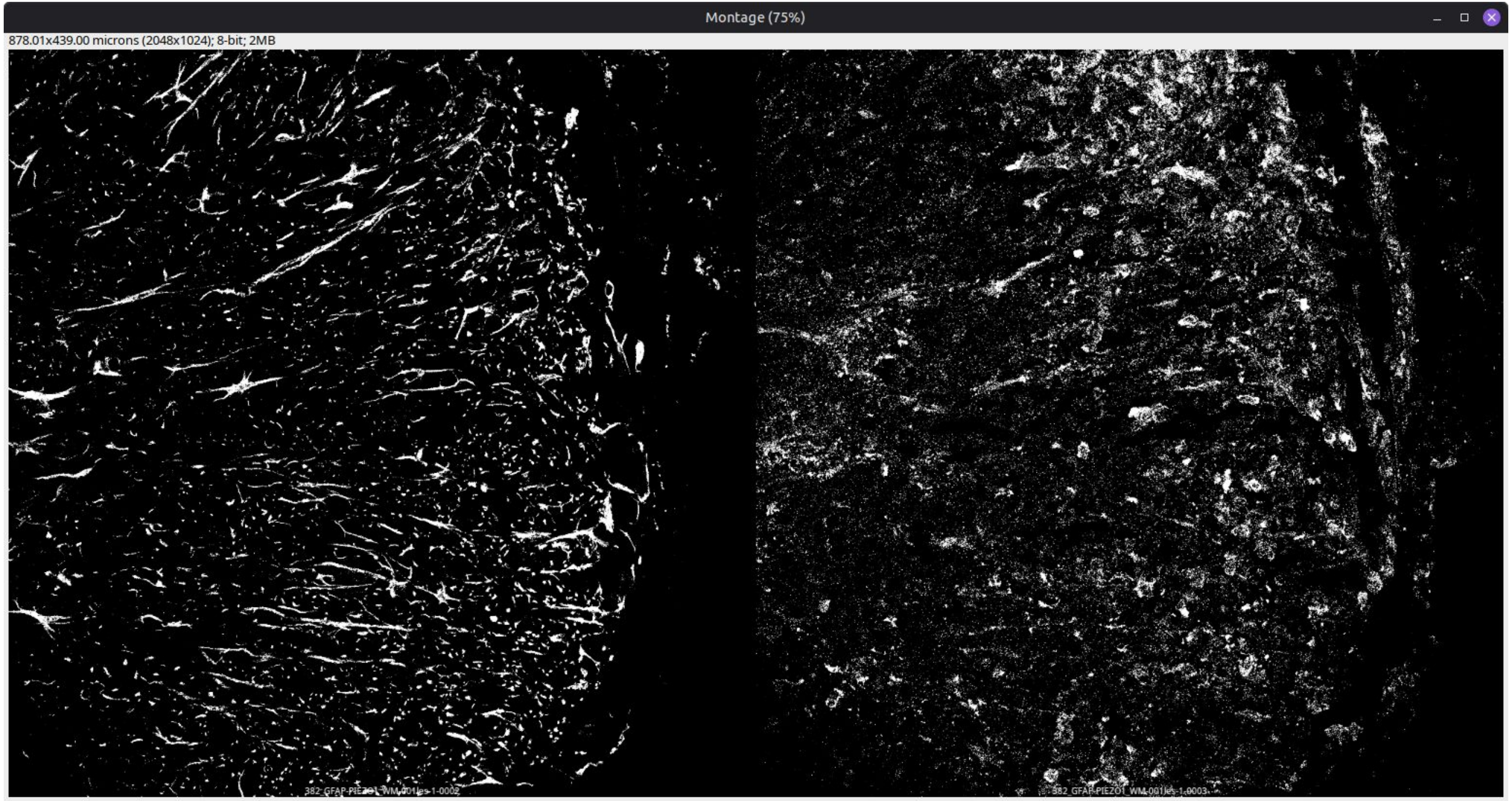


Split Channels 2,3

Downsample to 8-bit

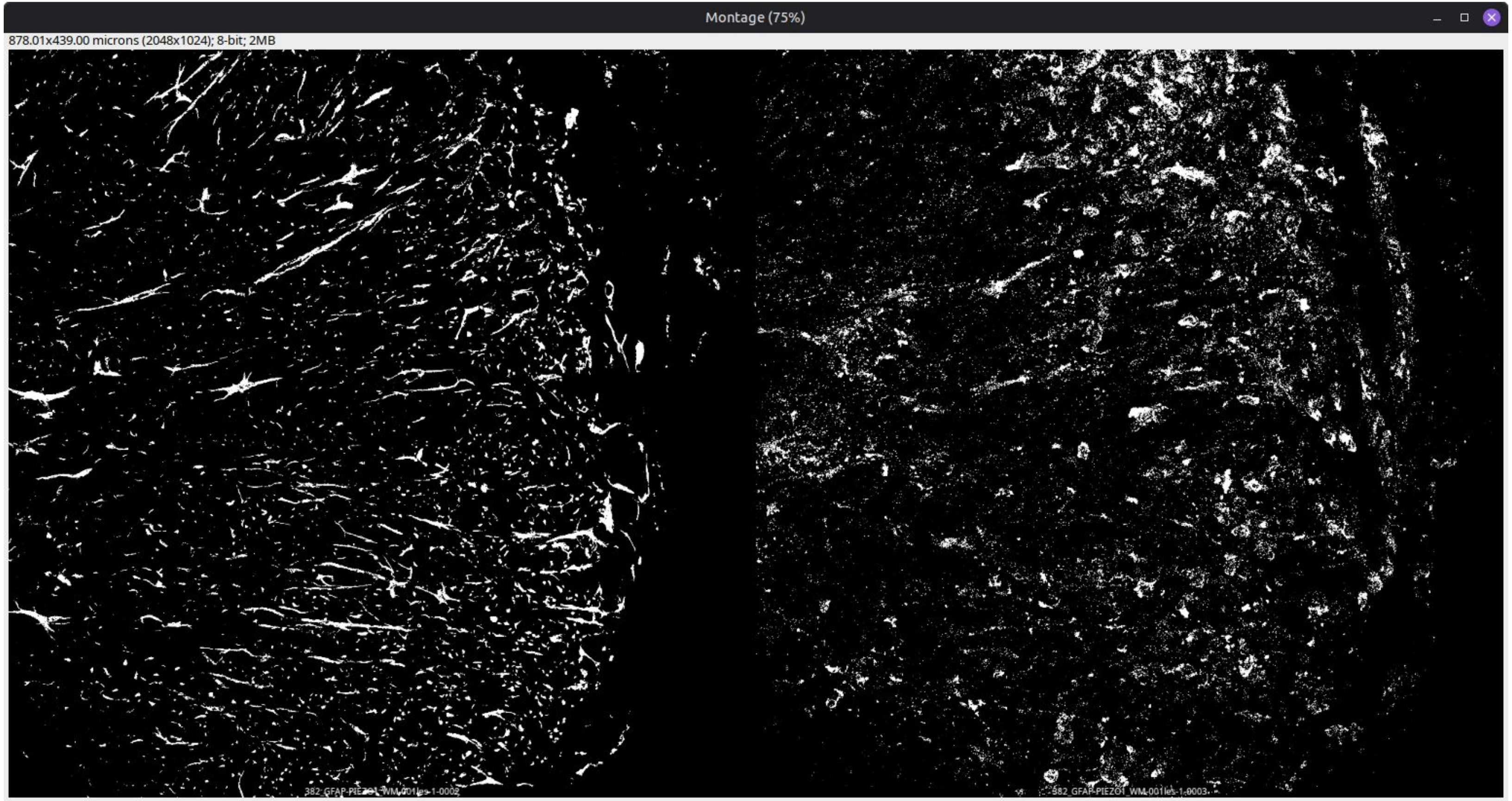


Processing 2/5 - Threshold



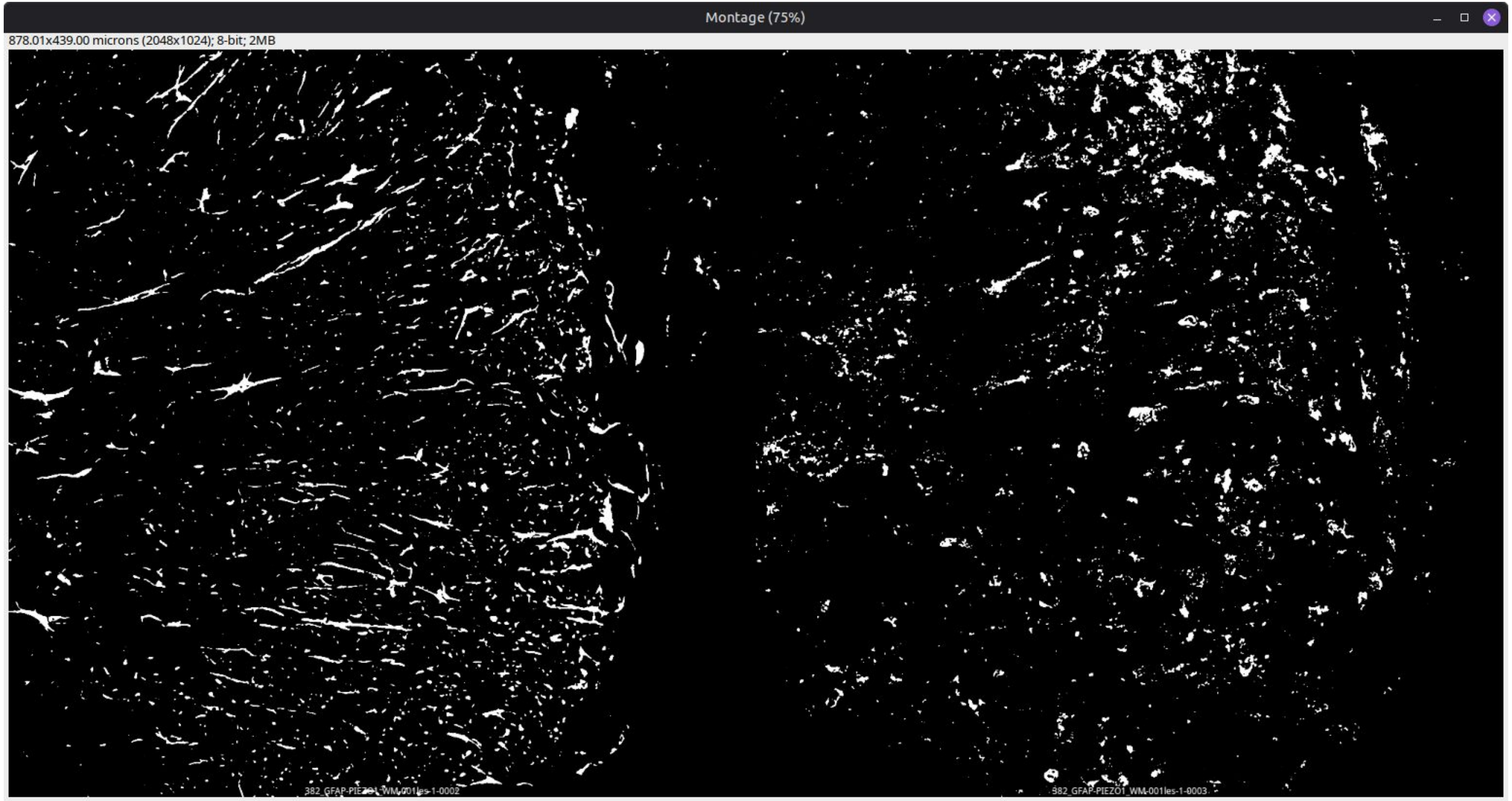
GFAP Piezo1

Processing 3/5 - Despeckle



GFAP Piezo1

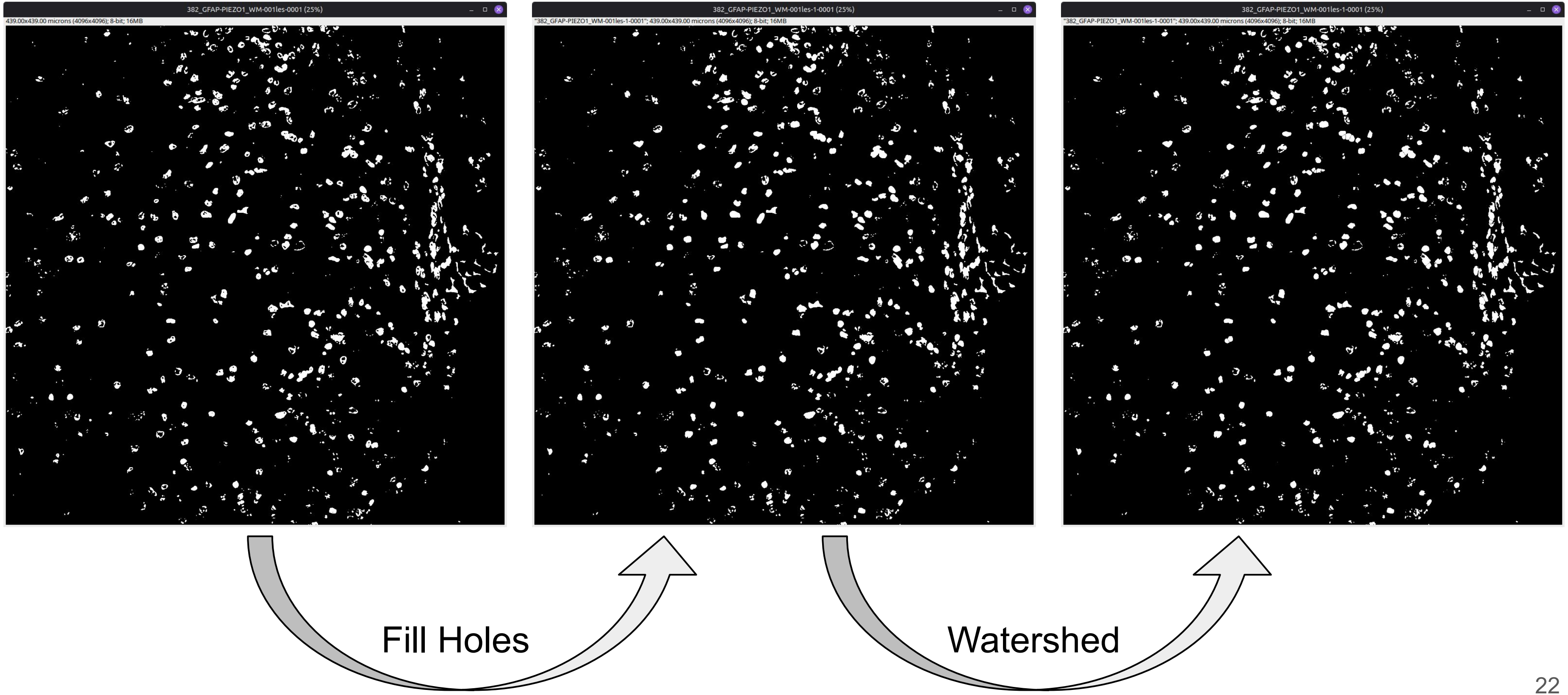
Processing 4/5 - Median Filter



GFAP Piezo1

Processing 5/5 - Channel 1 (Optional)

Is there any Ch1 Processing at all? If yes, and all previous steps have been completed



What's next?

The estimated thresholds can be used in other macros or plugins requiring pre-defined values for image segmentation

References

1. Schindelin, J., Arganda-Carreras, I., Frise, E. *et al.* **Fiji: an open-source platform for biological-image analysis**. *Nat Methods* 9, 676–682 (2012)
2. Dima, A.A., Elliott, J.T., Filliben, J.J. *et al.* **Comparison of segmentation algorithms for fluorescence microscopy images of cells**. *Cytometry*, 79A: 545-559 (2011)